

The 6-Month CTO Roadmap: Project Ava (Web MVP)

We will break this down into three major phases: **The Skeleton** (UI & Maps), **The Brain** (MaaS & AI), and **The Launch** (Polish & Deployment).

Phase 1: The Skeleton (Months 1 & 2)

Goal: Build the visual shell, user onboarding, and basic A-to-B mapping.

Timeline	Focus Area	Key Deliverables & Tech Stack
Month 1	Foundation & UI	Stack: Next.js, Tailwind CSS, Vercel. - Initialize the Next.js repository. - Build the wireframe onboarding flow: "Get Started," "Setup Preferences," and "Permissions." - Setup Firebase Authentication (Email/Google Sign-in).
Month 2	Mapping & Routing	Stack: Mapbox GL JS, Mapbox Directions API. - Integrate Mapbox to render the base map in the browser. - Build the Search Bar (Geocoding: turning "Times Square" into coordinates). - Implement basic A-to-B polyline drawing (standard routing).

Phase 2: The Brain (Months 3 & 4)

Goal: Upgrade the map into a true MaaS platform and inject the AI companion.

Timeline	Focus Area	Key Deliverables & Tech Stack
Month 3	MaaS Integration	Stack: Node.js backend APIs, GTFS parsers. • Build the "Journey Summary" and "Compare Modes" UI. • Connect mock/live APIs for Uber/Lyft pricing. • Ingest public transit data (GTFS) to show bus/train options alongside walking and rideshare.
Month 4	The Ava AI Engine	Stack: Google Gemini API. • Feed user preferences (e.g., "Eco-friendly," "Hates rain") into Gemini. • Intercept standard routing data and have the AI re-rank or suggest better multi-modal trips based on context. • Build the "AI Companion" chat interface for natural language trip planning.

Phase 3: The Launch (Months 5 & 6)

Goal: Tie the system together, handle the simulated booking, and deploy to the world.

Timeline	Focus Area	Key Deliverables & Tech Stack
Month 5	Booking & Polish	Stack: Stripe (Test Mode), React State Management. • Build the "Proceed to Pay" and "OTP Booking" screens from your wireframe. • Create the "Trip Summary" and "Booking Confirmed" post-trip loop. • Polish the UI for mobile-responsive browsers (Safari/Chrome).
Month 6	Deployment & Beta	Stack: Vercel (Frontend), Render/Heroku (Backend), Google Analytics. • Conduct end-to-end testing (QA). • Deploy the live web application. • Onboard 10-20 beta testers to try planning routes in their city. • Monitor logs and fix critical bugs.